

# CV Attendance System — Build Plan

## Architecture

## Project Structure

```
cv_attendance/  
├─ requirements.txt  
├─ config.py           # Global settings  
├─ database.py         # SQLite schema + CRUD  
├─ face_engine.py      # Detection, alignment, embedding  
├─ attendance_engine.py # State machine + attendance logic  
├─ register.py         # Student registration flow  
├─ camera.py           # Live camera + recognition loop  
├─ app.py              # Main entry point (CLI menu)  
└─ data/  
    └─ faces/          # Stored face images per student
```

## Database Schema

### students

| Column        | Type       | Description                     |
|---------------|------------|---------------------------------|
| id            | INTEGER PK | Auto-increment                  |
| name          | TEXT       | Student name                    |
| roll_number   | TEXT       | Roll number                     |
| class_name    | TEXT       | Class/section                   |
| face_encoding | BLOB       | Serialized 128-d face embedding |
| created_at    | TIMESTAMP  | Registration time               |

## attendance

| Column           | Type          | Description                        |
|------------------|---------------|------------------------------------|
| id               | INTEGER PK    | Auto-increment                     |
| student_id       | FK → students | Reference                          |
| date             | DATE          | Attendance date                    |
| class_name       | TEXT          | Class/section                      |
| first_seen       | TIMESTAMP     | First detection                    |
| last_seen        | TIMESTAMP     | Last detection                     |
| total_detections | INTEGER       | Detection count                    |
| status           | TEXT          | UNKNOWN/DETECTED/CONFIRMED/PRESENT |

## Attendance State Machine

UNKNOWN →(first detection)→ DETECTED →(3+ detections in 2 min)→ CONFIRMED →(seen over 5+ min span)→ PRESENT

- **Tolerates temporary disappearance** — student can vanish for up to 5 minutes without losing state
- **Movement tracking** — records first\_seen and last\_seen, ensuring the student stayed through class
- **Anti-spoof baseline** — requires consistent detections, not a single frame

## Tech Stack

- Python 3.10+
- OpenCV (camera, display)
- face\_recognition (dlib-based, 128-d embeddings)
- NumPy (embedding math)
- SQLite (zero-config database)

## Build Order

---

1. `config.py` — settings/thresholds
2. `database.py` — schema + CRUD operations
3. `face_engine.py` — detect, align, embed faces
4. `register.py` — registration flow
5. `attendance_engine.py` — state machine
6. `camera.py` — live feed + recognition
7. `app.py` — CLI entry point